Game Design Specification

CPT306 Assignment 1

Contents

Basic Information **1**

1 Design Choices and Rationale **2**
1.1 Numerical and Core Design Rationale 2
1.2 Core Design 2

2 Core Game Content Description **2**
2.1 Game Overview 2
2.2 Level Design (Detailed Description) 3
2.2.1 Level 1: Air Trial (Novice Tutorial Level) 3
2.2.2 Level 2: Extreme Ground-Air (Challenge Level) 3
2.3 Enemy Intelligent Pathfinding Mechanism 5
2.4 Operation Controls (Unified for All Levels) 5

3 Detailed Core Mechanisms **6**
3.1 Survival and Revival (Unified for All Levels) 6
3.2 Combat and Item System 6
3.3 Obstacle System (Unified Implementation, Configured by Level) 7
3.4 Interface and Data Management 7
3.5 Audio and Background Music System (Adjustable in Settings) 7

4 Complete Script List and Function Overview **7**

5 AI Tool Usage Instructions **8**

6 Running Instructions **9**

7 Known Issues **9**

Basic Information

- Unity Version: Unity 2022.3.12f1 LTS (Long-Term Support version, stable and compatible with 2D projects; the full version number meets the submission requirements for CPT306 assignment)
- Development Tools: Unity Editor 2022.3.12f1 + Visual Studio 2022
- Programming Language: C#

- Game Name: Ground-Air Dungeon Shooter

1 Design Choices and Rationale

1.1 Numerical and Core Design Rationale

Player Basic Settings: The maximum health (number of revives) is set to 10, balancing the novice fault tolerance in the first level and the challenge pressure in the second level.

Movement and Combat Values:

- Player Movement Speed: 3; Dash Speed: 15; Dash Cooldown: 1.5 seconds; Attack Cooldown: 0.3 seconds; Bullet Damage: 1. These values ensure smooth and comfortable air combat operations, adapting to the high-frequency bullet hell combat mode (parameters can be adjusted in the PlayerMovement script inspector).
- Level Progression Design: Strictly follows the "Novice Tutorial → Hardcore Advanced" logic. The first level is a basic training level, containing only a single enemy type and randomly generated obstacles; the second level introduces two enemy types and higher-density obstacles, significantly increasing the difficulty.
- Obstacle Design: Adopts a unified "destructible + indestructible" dual system, managed and generated by the MapGenerator script, with parameters configurable in the inspector.

1.2 Core Design

Enemy Design: The first level uses basic long-range aircraft; the second level is divided into long-range heavy fire aircraft and melee heavy tanks, fitting the ground-air combat theme. Among them, the long-range heavy fire aircraft is set to 10 points, with an attack interval of 2 seconds and an attack range of 5; the melee heavy tank has 2 health points and a relatively fast movement speed (higher than the basic speed of 2), which is easy to react to in a short time, and is set to 20 points (enemy-related parameters can be adjusted in the Enemy/AdvancedEnemy script inspector).

Obstacle System Adjustment: Both levels randomly generate destructible and indestructible walls according to the specified ratio. The first level has a low density (15 destructible, 15 indestructible), suitable for novices; the second level has a high density (40 destructible, 30 indestructible) to increase the challenge difficulty. The collision detection radius during generation is 0.45, and the maximum number of retries is 50 to ensure no overlapping obstacles.

Item System: The second level includes two core items (health packs, attack buffs). Health packs are divided into two types, with clear health deduction rules: the health lost by the player when dying is consistent with the damage value inflicted on enemies, complying with combat logic. The visual prompt for the duration of the attack buff item needs optimization.

2 Core Game Content Description

2.1 Game Overview

Theme: 2D Ground-Air Dungeon Combat. Players control a fighter jet in a closed 23×23 grid dungeon, avoiding and destroying enemies and obstacles.

Core Goal: Clear two levels. The first level helps players familiarize themselves with game operations and mechanisms through gameplay; the second level requires surviving and achieving

high scores under the pressure of two enemy types and high-density obstacles.

Map Specifications: 23×23 grid (map size can be set to 11.5 in the MapGenerator script inspector), with an indestructible boundary on the outer layer; destructible/indestructible obstacles are randomly generated inside according to the level difference ratio.

2.2 Level Design (Detailed Description)

2.2.1 Level 1: Air Trial (Novice Tutorial Level)

Level Purpose: Novice tutorial, core goal is to allow players to master flight operations, shooting logic, and basic survival skills under low pressure.

Clear Condition: Defeat all enemies within the 60-second countdown, then click the "Next Level" button on the transition page to enter the second level.

Failure Condition: The player exhausts 10 revives, triggering the failure screen.

Player Settings:

- Initial Maximum Health (Number of Revives): 10 (the "Maximum Health" parameter can be set in the PlayerMovement script inspector). After death, revive with full health at the designated revive point, no penalty.
- All basic operations (movement, attack, dash, pause) are unlocked for players to familiarize themselves with the tutorial.

Enemy Settings:

- Only one enemy type (Basic Fire Aircraft): A total of 10, generated all at once initially, no respawn mechanism.
- Enemy Attributes: Speed 2, Health 1, Score 10; only basic straight-line pursuit, no long-range attack; simple AI (parameters can be adjusted in the Enemy script inspector).

Map and Obstacles:

- Destructible Walls: 10 types, 15 in total; Indestructible Walls: 9 types, 15 in total. Low density and sparse distribution.
- Generation Rules: Randomly generated according to the inspector ratio; low-density design provides sufficient movement and shooting space, with generation parameters managed by the MapGenerator script.

Core Tutorial Goals:

- Master WASD movement, left mouse button shooting, Shift dash, and spacebar pause operations.
- Understand enemy pursuit, collision damage, and revival mechanisms.
- Control the player to rotate towards the mouse direction to achieve precise shooting.

2.2.2 Level 2: Extreme Ground-Air (Challenge Level)

Level Purpose: Advanced challenge, core goal is to test the player's precise operation, resource management, and multi-target handling capabilities.

Clear Condition: Defeat all generated enemies within the 60-second countdown to trigger the final victory screen.

Failure Condition: The player's health (number of revives) is exhausted, and the game fails.

Player Settings:

- Health (number of revives) is completely reset to 10 when entering the level to ensure fair challenge; the maximum health can be adjusted through the PlayerMovement script inspector.
- All operations from the first level are retained, with the addition of item collection strategies.

Enemy Settings (Two Types):

A total of 20 enemies. 10 are generated initially; after defeating one enemy, another enemy of the same type respawns after a 2-second delay until all 20 enemies are generated. Details are as follows:

1. Long-Range Heavy Fire Aircraft Core Features: High damage, long-range, fragile, dual muzzles, high threat.

Attributes (Adjustable in Inspector): Speed 2, Health 1, Score 10, Attack Interval 2 seconds, Attack Range 5.

Behavior Logic: Maintains distance, dual left and right muzzles (fires two bullets simultaneously, 1 damage per bullet); has obstacle avoidance function, prioritizes long-range output. When the player enters its attack range (Attack Range 5) and there are no obstacles blocking the line of sight between the two, the enemy will stop moving and fire bullets synchronously from two launch points at a fixed cooldown interval (Attack Interval 2 seconds) to attack the player; if the player is outside the attack range, the enemy will enter chase mode and execute intelligent pathfinding logic to approach the player; when colliding with the player, this long-range enemy will temporarily stop moving to avoid getting stuck in the player; after the enemy is defeated, it will randomly drop items according to the configured probability (20% Small Health Pack, 30% Large Health Pack, 50% Attack Buff), and trigger a global death event to notify other game systems.

2. Melee Heavy Tank Core Features: High health, high durability, melee charge, relatively fast movement, high threat.

Attributes (Adjustable in Inspector): Speed 3, Health 2, Score 20, attacks by charging at enemies.

Behavior Logic: Straight-line pursuit; high health requires multiple attacks to destroy; uses size and speed to block player movement. Since it has no shooting ability, this enemy is always in chase mode and executes intelligent pathfinding logic to approach the player; when colliding with the player, this melee enemy will directly deal 2 damage to the player and then die itself; after the enemy is defeated, it will randomly drop items according to the configured probability (20% Small Health Pack, 30% Large Health Pack, 50% Attack Buff), and trigger a global death event to notify other game systems.

Map and Obstacles:

- Destructible Walls: 10 types, 40 in total; Indestructible Walls: 9 types, 30 in total (the number can be adjusted in the MapGenerator script inspector).
- Generation Rules: High-density random generation according to inspector configuration; collision detection radius 0.45, maximum number of retries 50, controlled by the MapGenerator

script to avoid overlapping obstacles.

Difficulty: The number of indestructible walls increases, compressing space, forcing players to move flexibly in dense obstacles, avoiding long-range fire and tank charges. Players can use obstacles and dashes to dodge bullets.

Core Challenge Goals:

- Prioritize eliminating long-range enemies; use obstacles to block heavy tanks.
- Use items reasonably: Collect small/large health packs to restore health; use attack buff items to improve output efficiency against high-health targets.
- Precise Movement: Use the dash function to dodge attacks in dense obstacles. The dash speed is 15 with a 1.5-second cooldown, which can be adjusted in the PlayerMovement script inspector.

2.3 Enemy Intelligent Pathfinding Mechanism

The intelligent pathfinding of all enemies is centered on chasing the player, and they will always keep facing the player's direction. At the same time, they switch movement logic according to the distance from the player, line of sight, and their own state, adopting a three-stage pathfinding mechanism to achieve efficient enemy chasing and obstacle avoidance. Details are as follows:

1. Stuck Detection and Escape: First, detect whether it is in a stuck state (i.e., there is movement speed but minimal actual displacement). If judged to be stuck, it will immediately trigger escape mode, moving to the side away from the player to try to get out of the predicament of being blocked by obstacles and avoid long-term stagnation.

2. Multi-Direction Candidate Detection: In normal chase state, the enemy will simultaneously detect multiple candidate movement directions, including the positive direction towards the player, left and right offset directions, and wall-sliding directions when encountering obstacles, ensuring coverage of all possible paths to approach the player and avoid obstacles.

3. Optimal Path Selection: Through a built-in scoring mechanism, evaluate all candidate movement directions and select the optimal path that is "closest to the direction of the player" and "no obstacles blocking", so as to achieve precise obstacle avoidance and efficient enemy chasing, balancing pathfinding efficiency and chasing goals.

Supplementary Note: When the long-range heavy fire aircraft enters the attack range and has no line of sight obstruction, it will pause pathfinding and stop moving, focusing on firing bullets to attack the player; since the melee heavy tank has no shooting ability, it will continuously execute the above pathfinding logic until it collides with the player.

2.4 Operation Controls (Unified for All Levels)

Operation Controls: The specific operation methods, keys, and descriptions are as follows; core parameters can be adjusted in the corresponding script inspector.

1. Flight Movement Keys: W/A/S/D; Description: Move up, down, left, and right within the 23×23 grid; has collision detection, cannot pass through walls. During combat, the aircraft rotates towards the mouse direction to achieve precise shooting. The movement speed of 3 can be adjusted in the PlayerMovement script.

2. Attack Key: Left Mouse Button; Description: Fire bullets; cooldown time 0.3 seconds (adjustable); direction is consistent with the mouse aiming direction. Can cause damage to enemies and destroy destructible walls, with a basic bullet damage of 1.

3. Dash Key: Left Shift; Description: Temporarily increase speed to 15; cooldown time 1.5 seconds (both adjustable), used for emergency evasion.

4. Pause Key: Spacebar; Description: Pause the game; when paused, you can check the status, restart the level, or return to the main menu.

3 Detailed Core Mechanisms

3.1 Survival and Revival (Unified for All Levels)

Player Maximum Health (Number of Revives): 10 (can be adjusted in the PlayerMovement script inspector). The consumption of the player's health (number of revives) is consistent with the enemy's attack value (one bullet hit and one successful collision both deal 1 damage). The player triggers fixed-point revival when hit.

Revival Rules: Consume 1 revive count to revive with full health at the designated location. The game ends when all revive counts are exhausted.

Level Transition: Automatically enter the second level after clearing the first level; health resets to 10 when entering the second level.

3.2 Combat and Item System

Combat System:

- Player Bullets: Fired from the shooting point; basic damage 1 point, which can be temporarily increased by attack buff items. The relevant logic is implemented by the Bullet.cs and AttackBoostPickup.cs scripts.
- Enemy Attacks: Long-range heavy fire aircraft fire dual bullets (1 damage per bullet, 2 damage per attack), with an attack interval of 2 seconds and an attack range of 5 (adjustable); melee heavy tanks deal 2 damage through collision, with 2 health points (adjustable).
- Wall Interaction: Player bullets can destroy destructible walls; when hitting indestructible walls, they bounce or disappear, with wall attributes controlled by the MapGenerator script.

Item System (Only Available in Level 2):

- Small Health Pack: +1 Health, common.
- Large Health Pack: +2 Health, rare.
- Attack Buff: Temporarily increases damage by 1 point (total damage 2 points) for 5 seconds, which is effective against heavy tanks. The visual prompt for the duration of the attack buff item needs optimization, and the relevant logic is managed by the AttackBoostPickup.cs script.

3.3 Obstacle System (Unified Implementation, Configured by Level)

Two Types: Destructible Walls and Indestructible Walls. They share the same physical mechanism, with the number and density configured according to level differences, uniformly managed by the MapGenerator script, and all parameters can be adjusted in the inspector.

- Indestructible Walls: 9 types, cannot be destroyed. 15 in the first level; 30 in the second level, the number can be adjusted as needed.
- Destructible Walls: 10 types, can be destroyed by bullets. 15 in the first level; 40 in the second level, the number can be adjusted as needed.
- Generation Rules: Managed by the MapGenerator script; randomly generated within the 23×23 grid, avoiding the boundary; randomly rotated 0°–90°; collision detection radius 0.45, maximum number of retries 50, ensuring reasonable and non-overlapping generation.

3.4 Interface and Data Management

Real-Time Display Interface (HUD):

- Top Left Corner: Score, current health (remaining revives), 60-second countdown.
- Flow: Main Menu → Level 1 → Level 2 → Victory/Failure, managed by the SceneController and UIController scripts.
- Scoring Rules: Basic enemies in Level 1: 10 points; Long-range Heavy Fire Aircraft in Level 2: 10 points (dual bullet attack, easy to cause player health loss in a short time); Melee Heavy Tanks in Level 2: 20 points (2 health points, relatively fast movement, not easy to hit). The score difference in Level 2 mainly comes from the player's handling efficiency when facing melee heavy tanks: if the player fails to destroy the melee heavy tank in time, the tank will charge the player, causing the player to consume a revive count (respawn), and the tank will also be destroyed, but the player will not be able to obtain the 20 points corresponding to the melee heavy tank; if destroyed in time, the points can be obtained normally. Only the score of Level 2 is displayed finally, because all players have the same score in Level 1. Score statistics are implemented by the Enemy and LeaderboardReader scripts.
- Leaderboard: After winning Level 2, the player's score, name, and clear time are saved locally, managed by the LeaderboardReader script, and the data is stored in PlayerPrefs.

3.5 Audio and Background Music System (Adjustable in Settings)

Global Background Music (BGM) and Sound Effects (SFX) are centrally controlled by the AudioManager script; the settings interface can independently adjust the volume (0–100%) through a slider, and the volume settings are automatically saved with a default volume of 20%. Relevant parameters can be adjusted in the script inspector.

Core sound effects are divided into player shooting sound effects and hit sound effects, both synchronized with operational actions.

4 Complete Script List and Function Overview

All C# scripts used in the project and their core responsibilities:

1. AdvancedBullet.cs: Implements enhanced bullet logic (such as penetration, damage correction, special effects) for advanced combat mechanisms.

2. AdvancedEnemy.cs: Handles complex AI behaviors of advanced enemies (such as pathfinding, skill cooldown, obstacle avoidance), and enemy parameters can be adjusted in the inspector.

3. AttackBoostPickup.cs: Manages the collection of attack buff items, including trigger detection, temporary damage bonus application, and visual/audio feedback.

4. AudioManager.cs: Centralized audio control system, responsible for BGM, sound effects, and volume management, supporting global settings and playing sound effects according to actions. Volume parameters can be adjusted.

5. Bullet.cs: Core logic of player bullets, including movement, collision detection, damage calculation, and interaction with destructible walls.

6. Enemy.cs: Basic enemy class, implementing basic enemy behaviors (movement, pursuit, health, scoring), inherited by advanced enemy types. Basic enemy parameters can be adjusted.

7. EnemyBullet.cs: Implements enemy bullet logic, including movement, collision with players/obstacles, and damage application.

8. EnemySpawner.cs: Manages enemy spawning logic, including spawning timing, quantity, wave control, and level-specific enemy type configuration.

9. HealthPickup.cs: Handles health pack collection logic, including trigger detection, health recovery, and visual/audio feedback.

10. LeaderboardReader.cs: Manages leaderboard data, including local score saving, loading, sorting, and high-score interface display.

11. MapGenerator.cs: Core procedural map generation system, responsible for the random placement of destructible/indestructible obstacles, collision detection, and level-specific density configuration. All generation parameters can be adjusted in the inspector.

12. PlayerMovement.cs: Core player controller, handling movement, dash, rotation, shooting, health/revival, and input processing. Core player parameters can be adjusted in the inspector.

13. SceneController.cs: Manages game scene flow, including level transitions, game state control (pause/restart), and main menu navigation.

14. Temp.cs: A temporary test/debug script during development, used for logic verification and functional prototype development.

15. UIController.cs: Manages all in-game interface elements, including HUD display (health, score, timer), pause menu, victory/failure interface, and settings interface.

5 AI Tool Usage Instructions

- Whether AI Tools are Used: Yes
- Tool Name: Doubao AI
- Auxiliary Tasks: C# code framework reference, dual-enemy logic design, obstacle generation algorithm optimization, format typesetting and polishing of the instruction document.
- Statement: All core logic, implementation code, and numerical balance are original. AI is only used for reference and assistance, with no direct plagiarism, fully complying with academic integrity rules.

6 Running Instructions

- Platform: Windows 64-bit
- Running Method: Run the .exe file directly; ensure the file path contains no Chinese characters.
- Data: Leaderboard data is saved locally; if the data is corrupted, the PlayerPrefs file can be deleted to reset.

7 Known Issues

1. Even with obstacle avoidance enabled, when an obstacle is between the player and the enemy, the enemy's pathfinding may occasionally get lost.
2. The visual prompt for the duration of the attack buff item needs optimization to provide clearer feedback.
3. Only the score of Level 2 is displayed in the final leaderboard, not including the score of Level 1, because all players have the same score in Level 1.
4. Currently, the game can only be resumed by clicking "Resume" in the pause menu, not by pressing the spacebar again.
5. The game failure is triggered when the health is not 0 at the end of the game.